

Lecture #9: Linear Programming Problem

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A Lumber Yard

A lumber yard makes two products: finish-grade and construction grade boards.

- ① two hours to rough saw each one thousand board feet of the finish grade boards, five hours to plane each 1000 board feet of these boards
- ② two hours too rough saw each 1000 feet of construction grade boards but it takes only three hours to pay in each 1000 board feet of these boards
- ③ The saw is available eight hours per day and the plane is available 15 hours per day.

If the profit on each 1000 board feet of finish grade boards is \$120 and the profit on each 1000 board feet of construction grade board is \$100, how many board feet of each type of lumber should be sold to maximize profit?



Let f and c be the amount of finish-grade boards and construction-grade boards, respectively, measured in 1000 board feet.

For the saw is only available 8 hours, we have

For the plane, which is available 15 hours, we have

This means our math problem is: Find the values of f and c that will



A Nutritionist problem

A Nutritionist wants to make sure that you are getting your macros. They have two food options.

- ① Each ounce of A contains two units of fat, one unit of carbohydrates, and four units of protein.
- ② Each ounce of B contains three units of fat, three units of carbohydrates, and three units of protein.
- ③ The nutritionist wants the meal to provide at least 18 units of fat, at least 12 units of carbohydrates, and at least 24 units of protein.

If an ounce of A costs \$0.20 and an ounce of B costs \$0.25 how many ounces of each should be served to minimize the cost of the meal while satisfy the nutritionist requirements?



Let A and B be the amount of ounces of food A and B , respectively.
To meet the fat recommendation

To meet the carbohydrates and protein, we must have

Of course, we have $A \geq 0, B \geq 0$. The cost (in dollars) to be minimized is

This means our math problem is: Find the values of A and B that will



Compare and Constrast

Find the values of f and c that will
maximize $z = 120f + 100c$ subject to

$$\left. \begin{array}{rcl} 2f + 2c & \leq & 8 \\ 5f + 3c & \leq & 15 \\ f & \geq & 0 \\ c & \geq & 0. \end{array} \right\}$$

Find the values of A and B that will
minimize $z = 0.2A + 0.25B$ subject to

$$\left. \begin{array}{rcl} 2A + 2B & \geq & 18 \\ A + 3B & \geq & 12 \\ 4A + 3B & \geq & 24 \\ A & \geq & 0 \\ B & \geq & 0. \end{array} \right\}$$



General Form

Definition (General Linear Programming Problem)

The *general linear programming problem* can be stated as the following: Find values of $x_1, x_2, \dots, x_n$ that will

maximize or minimize $z = c_1x_1 + c_2x_2 + \dots + c_nx_n$ subject to the restrictions (1)

$$\left. \begin{array}{rcl} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n & \leq (\geq)(=) & b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n & \leq (\geq)(=) & b_2 \\ \vdots & & \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n & \leq (\geq)(=) & b_m \end{array} \right\} \quad (2)$$

where only one $\geq, \leq$ or $=$ appears. We call the linear function 1 the *objective function*. The (in)equalities in 2 are called the *constraints*.



Standard Form

Definition (Standard Programming Problem)

The *general linear programming problem* can be stated as the following: Find values of $x_1, x_2, \dots, x_n$ that will

maximize or minimize $z = c_1x_1 + c_2x_2 + \dots + c_nx_n$ subject to the restrictions

$$\left. \begin{array}{rcl} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n & \leq (\geq)(=) b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n & \leq (\geq)(=) b_2 \\ \vdots & & \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n & \leq (\geq)(=) b_m \end{array} \right\}$$
$$x_j \geq 0, \quad \text{for all } j = 1, 2, \dots, n$$



Canonical Form

Definition (Canonical Programming Problem)

The *general linear programming problem* can be stated as the following: Find values of $x_1, x_2, \dots, x_n$ that will

maximize or minimize $z = c_1x_1 + c_2x_2 + \dots + c_nx_n$ subject to the restrictions

$$\left. \begin{array}{rcl} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n & \leq (\geq)(=) b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n & \leq (\geq)(=) b_2 \\ \vdots & & \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n & \leq (\geq)(=) b_m \end{array} \right\}$$
$$x_j \geq 0, \quad \text{for all } j = 1, 2, \dots, n$$



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1 Getting into Standard Form

2 Matrix Equations



Going from Max to Min

Every minimization problem can be viewed as a maximization problem.

$$\min c_1x_1 + c_2x_2 + \cdots + c_nx_n =$$

If we multiply an inequality



From Equality to Inequality

To change an equality to an inequality we need to observe that $x = 6$ is the same as $x \leq 6$ and $x \geq 6$, or $-x \leq -6$. For a generic pair of inequalities,

$$\sum_{j=1}^n a_{ij}x_j = b_i,$$

we could get to



Unconstrained to Constrained

While we want all variables to be ≥ 0 , that is not always the case, so assume x_j is not constrained to be nonnegative.

Lastly as we talked about with linearity, if we scale one side of the constraints of a linear programming problem, then the optimal solution is the same as the adjusted problem.



Linear Programming Examples

Example (A Blending Problem)

A manufacturer of artificial sweetener blends 14 kg of saccharine and 18 kg of dextrose to prepare two new products: SWEET and LO-SUGAR. Each kilogram of SWEET contains 0.4 kg of dextrose and 0.2 kg of saccharin, whereas each kilogram of LO-SUGAR contains 0.3 kg of dextrose and 0.4 kg of saccharin. If the profit on each kilogram of SWEET is \$0.20 and the profit on each kilogram of LO-SUGAR is \$0.30 how many kilograms of each product should be made to maximize the profit?



Answer to a Blending Problem

Let x, y be the number of kg of SWEET and LO-SUGAR made respectively.



Financial Problem

Let x and y denote the amounts invested in bond AAA and stock BB, respectively. The university wants to maximize return, They have \$100,000 to invest, and they want to be risk averse, so they don't want to put too many eggs in one basket. Specifically AAA has to be more than 2 times BB, and BB has to be less than 30,000 total. AAA has a 7% return and BB has a 9% return.



That means Find values of x, y that will



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① Getting into Standard Form

② Matrix Equations



Consider the standard form of the linear programming problem:
Find values of $x_1, x_2, \dots, x_n$ that will

$$\textbf{minimize } z = c_1x_1 + c_2x_2 + \cdots c_nx_n \text{ subject to the restrictions} \quad (3)$$

$$\left. \begin{array}{rcl} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n & \leq & b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n & \leq & b_2 \\ \vdots & & \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n & \leq & b_m \end{array} \right\} \quad (4)$$

$$x_j \geq 0, \quad \text{for all } j = 1, 2, \dots, n \quad (5)$$



If we let

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}, \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \cdots \\ x_n \end{pmatrix}, \mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ \cdots \\ b_m \end{pmatrix}, \mathbf{c} = \begin{pmatrix} c_1 \\ c_2 \\ \cdots \\ c_n \end{pmatrix}$$

We can write this as Find the vector $\mathbf{x} \in \mathbb{R}^n$ that will

